Membership Domain API - Design

Overview

The Membership module provides a comprehensive set of RESTful APIs for managing membership data and related entities. It covers key areas such as policy details, covered individuals, alerts, notes, fulfilment, digital cards, communications, financial information, and organisational relationships. The module is designed for secure, efficient, and scalable integration with other systems.

Key Entities:

• Alert • Note • PolicyDetails • PersonCovered • PersonCoverHistory • PackageInfo • Fund • Badge • Brand • PremiumDetails • Rebate • LhcLoading • Abd	• CampaignOfferDetails • ClearanceRequest • CommunicationPreference • DefaultPostalAddress • TaxStatementSummary • Cover • Fulfilment • Cvo • DigitalCard • IncomeTier • Organisation • OrganisationSalesMessage

Technical Product Owner(s): Anbarasan Rajangam

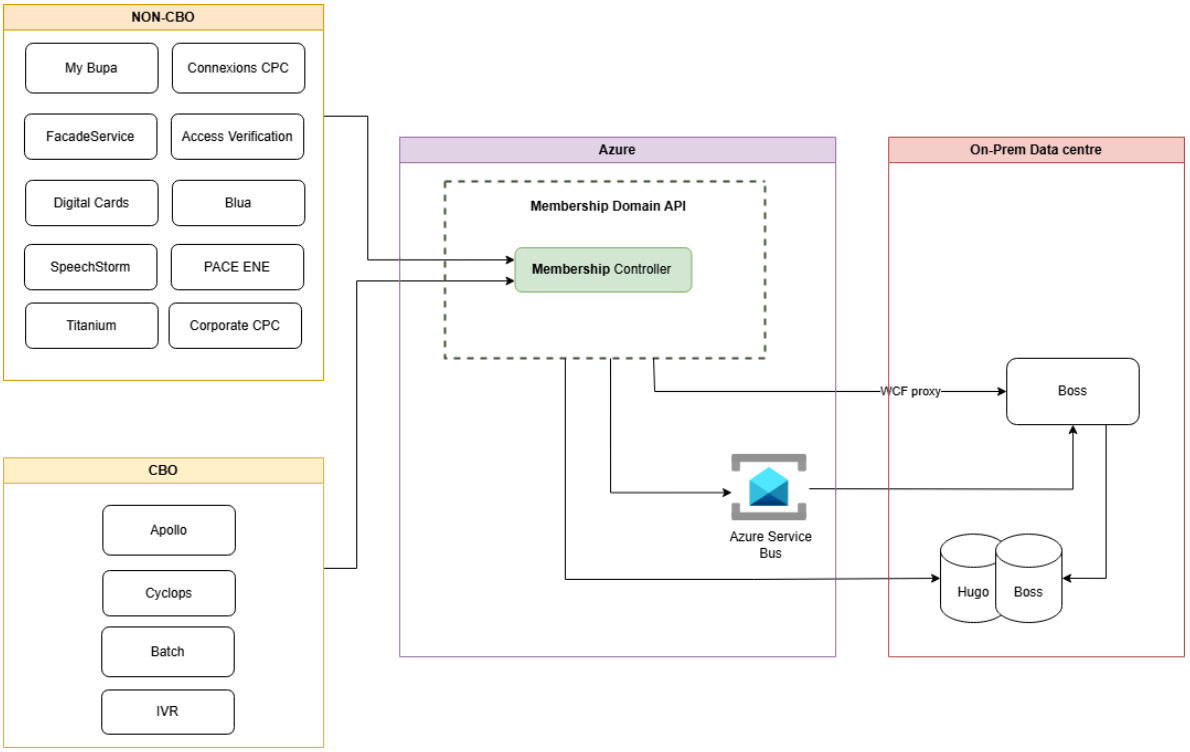
Stakeholders:

The key stakeholders are listed below, for the full list of project stakeholders refer [Project Stakeholders - DotNet Team - Confluence]

Upstream App name	Technical Owner
Apollo	Albert Gouzali / Andre Margono
Cyclops	Anbarasan Rajangam
Batch Jobs	Anbarasan Rajangam
Digital Cards	Andy Prince / Jonathan Sher
Digital (MyBupa)	Gabriel Lopez, Andy Prince
Facade Services	Anbarasan Rajangam
IVR	Rohit Suvarna
SpeechStorm	Rohit Suvarna
Titanium	Goran Fak/Gautam Bhandarkar
Corporate CPC	Antony Sideropoulos

Connexions CPC	Hemalatha Paramasivam/Antony Sideropoulos
Access Verification	Jonathan Sher
Blua	Cameron Biggelaar
PACE ENE	Brian Morrison

High-Level Integration Diagram



Domain Model

```
{
    "membershipId": "long",
    "isMembershipPorted": "bool",
    "enrollmentYear": "int",
    "summaryAsOf": "DateTime",
    "businessIndicator": "string",
    "sourceMembershipId": "string",
    "hasSpe": "bool",
    "speCode": "int",
    "lapsedRiskIndicator": "string",
    "lapsedIndicatorType": "string",
    "status": {
        "status": "int",
        "membershipStatusStartDate": "DateTime",
        "reactivationDate": "DateTime?",
        "deactivationDate": "DateTime?",
        "deactivatedBy": "string",
        "deactivationReason": "string",
        "isFutureSuspended": "bool"
    },
    "group": {
        "groupType": "string",
        "groupDatePaidTo": "DateTime?",
        "group": "string",
        "paymentGroupOrgDiscount": "double"
    },
    "alerts": [
        {
            "alertId": "long",
            "about": "string",
```

```
            "alertType": "enum | Unknown, System, Person, Membership, Session, BusinessValidationErrors, CSSMembership, CSSPerson, Dynamic",
            "severity": "enum | Unknown, Information, Warning, Error",
            "expiryTrigger": "enum | Maintainable, Validation, Session",
            "categoryName": "string",
            "action": "string",
            "text": "string",
            "userId": "string",
            "categoryId": "long?",
            "groupId": "long?",
            "groupName": "string",
            "policyId": "long?",
            "personId": "long?",
            "isNoteAttached": "bool",
            "persistedDynamicAlert": "bool",
            "dateCreated": "DateTime"
        }
    ],
    "notes": [
        {
            "text": "string",
            "categoryId": "long?",
            "personId": "long?",
            "policyType": "long?",
            "policyId": "long?",
            "customerInfoId": "long",
            "textId": "long?"
        }
    ],
    "personsCovered": [
        {
            "personId": "int",
            "firstName": "string",
            "lastName": "string",
            "suffix": "int",
            "gender": "string",
            "dateOfBirth": "Date",
            "dateDeceased": "Date?",
            "personsCoveredHistory": [
                {
                    "relationshipId": "int",
                    "relationshipDesc": "string",
                    "status": "string",
                    "personMembershipId": "long?",
                    "roleTimestamp": "DateTime?",
                    "membershipTimestamp": "DateTime?",
                    "startDate": "Date",
                    "endDate": "Date?"
                }
            ]
        }
    ],
    "coverDetails": [
        {
            "fromDate": "datetime",
            "toDate": "datetime",
            "status": "string",
            "state": "string",
            "familyType": {
                "familyTypeId": "int",
                "familyTypeDesc": "string"
            },
            "brand": {
                "id": "Int64",
                "name": "string",
                "abbreviation": "string"
            },
            "badge": {
                "id": "Int64",
                "name": "string",
                "abbreviation": "string",
                "isReinsurance": "bool"
            },
            "fund": {
                "id": "Int64",
                "name": "string",
                "abbreviation": "string",
                "state": "State"
            },
            "packages": {
                "packageCode": "string",
                "rateRebate": "decimal",
                "rate": "decimal",
                "discount": "decimal",
                "lifeHealthCoverAmount": "decimal",
```

```
                "gstAmount": "decimal",
                "riskLoadingAmount": "decimal?",
                "excess": "bool",
                "coPay": "bool",
                "excessBonus": "bool",
                "startDate": "DateTime",
                "endDate": "DateTime",
                "abdAmountDetails": {
                    "abDAmount": "decimal",
                    "abdAdult1Amount": "decimal",
                    "abdAdult2Amount": "decimal"
                },
                "hospitalProduct": {
                    "productOptionID": "string",
                    "productDescription": "string",
                    "productType": "string"
                },
                "ancillaryProduct": {
                    "productOptionID": "string",
                    "productDescription": "string",
                    "productType": "string"
                },
                "nationalPharmacyProduct": {
                    "productOptionID": "string",
                    "productDescription": "string",
                    "productType": "string"
                },
                "familyType": {
                    "familyTypeId": "int",
                    "familyTypeDesc": "string"
                }
            }
        }
    ],
    "lhcLoading": {
        "membershipId": "long",
        "adultOneCEA": "CertifiedEntryAge",
        "adultTwoCEA": "CertifiedEntryAge",
        "adultOneAbsentDays": "int",
        "adultTwoAbsentDays": "int",
        "adultOneEffectiveEntryAge": "int",
        "adultTwoEffectiveEntryAge": "int",
        "adultOneLoading": "double",
        "adultTwoLoading": "double",
        "lhcOverrideAdult1": "bool",
        "lhcOverrideAdult2": "bool",
        "membershipLoading": "double",
        "adultOneRelationType": "Relation",
        "adultTwoRelationType": "Relation",
        "adultOneConfirmStatus": "string",
        "adultTwoConfirmStatus": "string",
        "adultOneFundCode": "string",
        "adultTwoFundCode": "string",
        "adultOneHospitalUpToDate": "DateTime",
        "adultTwoHospitalUpToDate": "DateTime",
        "adultOneCertificateIssueDate": "DateTime?",
        "adultTwoCertificateIssueDate": "DateTime?"
    },
    "abdLoading": {
        "abdPercentage": "double",
        "rebatePercentage": "double",
        "abdAdult1Percentage": "double",
        "abdAdult2Percentage": "double",
        "discountPercentage": "double",
        "lhcPercentage": "double",
        "lhcAdult1Percentage": "double",
        "lhcAdult2Percentage": "double",
        "membershipLoading": "double",
        "adult1Loading": "double",
        "adult2Loading": "double",
        "adult1CDA": "CertifiedDiscountAge",
        "adult2CDA": "CertifiedDiscountAge",
        "abdFlag": "string",
        "adultOneHospitalUpToDate": "DateTime",
        "adultTwoHospitalUpToDate": "DateTime",
        "adultOneCertificateIssueDate": "DateTime?",
        "adultTwoCertificateIssueDate": "DateTime?",
        "adultOneABDConfirmStatus": "bool?",
        "adultTwoABDConfirmStatus": "bool?",
        "riskLoadingAdult1Percentage": "double?",
        "riskLoadingAdult2Percentage": "double?",
        "riskLoadingPercentage": "double?"
    },
    "organisation": {
        "id": "long?",
        "description": "string",
        "organisationType": "string",
        "discountPercentage": "long?"
```

```
    },
    "availability": {
        "ancillaryClaimsAvailable": "bool",
        "ancillaryBenefitsAvailable": "bool",
        "ancillaryTopUpAvailable": "bool",
        "hospitalClaimsAvailable": "bool"
    },
    "characteristics": {
        "characteristicType": {
            "type": {
                "id": "int",
                "description": "string",
                "flag": "string"
            },
            "rank": "int",
            "updated": "DateTime",
            "updatedBy": "string"
        }
    },
    "paymentInfo": {
        "paymentType": "string",
        "paymentFrequency": {
            "noun": "string"
        },
        "renewalFrequency": "string",
        "premium": "decimal?",
        "lastRenewalSentDate": "DateTime?",
        "premiumDateTo": "DateTime?",
        "premiumDateFrom": "DateTime?",
        "lastPaymentAmount": "decimal",
        "lastSuccessfulPayDate": "DateTime?",
        "nextPaymentProcessDate": "DateTime?",
        "futurePaymentRequestDate": "DateTime?",
        "paymentMethod": "string",
        "datePaidTo": "DateTime"
    },
    "clearances": {
        "incoming": [
            {
                "personId": "long",
                "clearanceInPersonRequest": {
                    "personExtFundAuthoritId": "long?",
                    "personId": "long",
                    "memberNo": "long",
                    "authority": "bool",
                    "previousFund": "string",
                    "previousFundNumber": "string",
                    "previousFundCardName": "string",
                    "bupaJoinDate": "DateTime",
                    "createDate": "DateTime?",
                    "auditTimestamp": "DateTime",
                    "mail1Date": "DateTime?",
                    "mail2Date": "DateTime?",
                    "seekAuthorityDate": "DateTime?",
                    "revokationDate": "DateTime?",
                    "receievedDate": "DateTime?",
                    "cancelledDate": "DateTime?",
                    "auditId": "long",
                    "portalIndicator": "bool",
                    "aliasFirstName": "string",
                    "aliasLastName": "string",
                    "upi": "int?",
                    "transactionId": "string",
                    "statusMsg": "string",
                    "ccpStatusDate": "DateTime?",
                    "productTypeId": "long?",
                    "history": [
                        {
                            "transactionId": "string",
                            "referenceId": "string",
                            "channel": "string",
                            "processDate": "DateTime?",
                            "message": "string",
                            "status": "string",
                            "code": "string",
                            "responseText": "string",
                            "userId": "string"
                        }
                    ]
                }
            }
        ],
        "outgoing": [
            {
                "messageType": "string",
                "transactionId": "string",
                "memberNo": "string",
                "transactionDateTime": "DateTime?",
```

```
                "eventDate": "DateTime?",
                "externalFundName": "string"
            }
        ]
    },
    "membershipCommPreferences": [
        {
            "id": "long",
            "personId": "long?",
            "communicationPreferenceTypeId": "int",
            "communicationPreferenceTypeDesc": "string",
            "hasAuthority": "bool",
            "detail": "string",
            "display": "string",
            "contactTypeId": "int",
            "contactTypeDesc": "string",
            "isValid": "bool",
            "verifiedDate": "DateTime?",
            "locatorTypeId": "long?",
            "inferredContactId": "long?",
            "hasValidationOverride": "bool",
            "country": "Country",
            "isUnknown": "bool",
            "isOptOut": "bool",
            "isOverride": "bool",
            "CommunicationCategoryType": "string",
            "auditId": "long"
        }
    ],
    "communications": [
        {
            "id": "long",
            "categoryId": "long",
            "categoryDescription": "string",
            "communicationItemId": "long",
            "communicationItemDescription": "string",
            "communicationDetail": "string",
            "requestedDate": "DateTime?",
            "generatedDate": "DateTime?",
            "processedDate": "DateTime",
            "person": "string",
            "recepient": "string",
            "channelId": "long",
            "channelDescription": "string",
            "statusId": "long",
            "statusDescription": "string",
            "hasError": "bool",
            "referenceId": "string",
            "hasRelatedCommunication": "bool",
            "isResent": "bool",
            "isPrinted": "bool",
            "isAllowedToResend": "bool",
            "isAllowedToPrint": "bool",
            "history": [
                {
                    "id": "long",
                    "processedDate": "DateTime",
                    "statusId": "long",
                    "statusDescription": "string",
                    "channelId": "long",
                    "channelDescription": "string",
                    "contactDetail": "string",
                    "user": "string",
                    "retailCentre": "string",
                    "isBundled": "bool?",
                    "hasError": "bool",
                    "errorDescription": "string",
                    "comments": "string",
                    "isLatestCommunicationHistory": "bool",
                    "documents": [
                        {
                            "mimeType": "MimeType",
                            "renderingServiceType": "RenderingServiceType",
                            "documentId": "long?",
                            "documentDescription": "string",
                            "fileName": "string",
                            "fileSize": "string",
                            "renderingTemplateId": "long?",
                            "renderingTemplateName": "string",
                            "isPrinted": "bool",
                            "isContentAvailable": "bool",
                            "fileLocation": "string"
                        }
                    ]
                }
            ]
        }
```

```
    ],
    "taxStatements": [
        {
            "personId": "long",
            "personName": "string",
            "referenceId": "long",
            "taxYear": "int",
            "type": "string",
            "status": "string",
            "dateRequested": "DateTime",
            "userId": "string",
            "contact": {
                "detail": "string",
                "display": "string",
                "contactTypeId": "int",
                "contactTypeDesc": "string",
                "isValid": "bool"
            },
            "bundled": "bool",
            "statementType": "string",
            "dateProcessed": "DateTime?",
            "mailingIsInvalid": "bool",
            "taxStatementDetail": [
                {
                    "paid": "int",
                    "rebateReceived": "int",
                    "rebateCode": "string",
                    "coPhiibName": "string",
                    "sequence": "long",
                    "coPhiibSuffix": "int?"
                }
            ],
            "personCoverTaxEnquiry": [
                {
                    "personName": "string",
                    "relationship": "enum | Unknown, Contributor, Partner, Daughter, Son, Husband, Wife, InterestedParty",
                    "startDate": "DateTime",
                    "endDate": "DateTime",
                    "age": "int",
                    "suffix": "int"
                }
            ]
        }
    ],
    "flags": [
        {
            "id": "long",
            "auditId": "long",
            "entityType": "enum | M",
            "applicableDate": "Date",
            "auditTimeStamp": "DateTime?",
            "entityFlagSelectedId": "long?",
            "flagValue": "int?",
            "startDate": "DateTime",
            "endDate": "DateTime"
        }
    ],
    "digitalCards": [
        {
            "cardNumber": "string",
            "cardType": "enum | Plastic, Digital",
            "digitalCardTypeId": "long",
            "digitalCardTypeName": "long",
            "digitalCardStatusId": "long",
            "digitalCardStatusName": "long",
            "startDate": "date",
            "endDate": "date"
        }
    ],
    "physicalCards": [
        {
            "currentFund": "string",
            "currentBrand": "string",
            "name": "string",
            "cardNumber": "string",
            "address": "string",
            "status": "string",
            "hasProductName": "bool",
            "usesAltAddress": "bool",
            "requestedDate": "DateTime",
            "issuedDate": "DateTime?",
            "cancelReason": "string",
            "cancelDate": "DateTime?",
            "suffix": "int?",
            "cardType": "enum | Plastic, Digital"
        }
    ],
```

```
    "incomeTier": [
        {
            "incomeTierLevelId": "int",
            "incomeTierDescription": "string",
            "financialYear": "string",
            "startDate": "Date",
            "endDate": "Date"
        }
    ],
    "internationalPartners": [
        {
            "internationalPartnerId": "long",
            "description": "string",
            "customerId": "string",
            "startDate": "DateTime",
            "endDate": "DateTime?",
            "sessionId": "long",
            "personId": "long",
            "countryId": "string",
            "country": "string"
        }
    ],
    "riskIndicator": {
        "riskIndicator": "string",
        "lapsedIndicatorType": "string"
    },
    "modalities": [
        {
            "id": "string",
            "description": "string",
            "seqId": "int",
            "parentId": "int?",
            "isExcluded": "bool",
            "serviceDescription": "string",
            "sortOrder": "int",
            "topModality": "int?",
            "startDate": "DateTime?",
            "endDate": "DateTime?",
            "maintainOptions": {
                "Change": "bool",
                "Replace": "bool",
                "Delete": "bool",
                "ApplicationError": "bool",
                "ChangeNextYear": "bool"
            },
            "maintainReasons": [
                "string"
            ],
            "claimGroup": "int"
        }
    ],
    "membershipCorporateSubsidy": {
        "membershipNumber": "long",
        "startDate": "DateTime",
        "endDate": "DateTime",
        "baseAmount": "decimal?",
        "subsidyAmount": "decimal",
        "percentage": "decimal?",
        "limitValue": "decimal",
        "familyType": "string",
        "paymentDate": "DateTime",
        "rateDate": "DateTime",
        "subsidyType": "string",
        "packageCode": "string",
        "excludeRebate": "bool",
        "excludeABD": "bool",
        "excludeLHC": "bool",
        "subsidyTypeCode": "string",
        "amountPerMonth": "decimal",
        "limitPerMonth": "decimal",
        "fundNumber": "int",
        "subsidyLevelOfCover": "string",
        "total": {
            "totalAmount": "decimal",
            "totalDiscount": "decimal"
        }
    }
}
```

Non-Restful Model

```
{
    "contacts": [
        {
            "id": "long",
```

```
            "personId": "long",
            "contact": {
                "contactTypeId": "long",
                "contactDetail": "string",
                "countryId": "string",
                "isValid": "bool",
                "hasValidationOverride": "bool",
                "isUnknown": "bool",
                "locatorTypeId": "long",
                "inferredContactId": "long",
                "verifiedDate": "DateTime?"
            },
            "address": {
                "addressee": "string",
                "addressLine1": "string",
                "addressLine2": "string",
                "addressLine3": "string",
                "postcode": "string",
                "locality": "string",
                "state": "string",
                "latitude": "decimal?",
                "longitude": "decimal?",
                "reliability": "int?",
                "dpid": "string?",
                "postalType": "string?",
                "gnafId": "string?"
            },
            "startDate": "DateTime",
            "endDate": "DateTime",
            "auditId": "long",
            "auditIdNew": "int?",
            "auditTimestampUtc": "DateTime?"
        }
    ],
    "cvo": [
        {
            "id": "int",
            "cvoId": "Guid",
            "name": "string",
            "startDate": "Date",
            "endDate": "Date",
            "displayOrder": "int",
            "hasRedemptionLimit": "bool",
            "requestLimit": "int",
            "certificateName": "string",
            "logonUrl": "string",
            "registrationUrl": "string",
            "pingUrl": "string"
        }
    ],
    "membershipDecode": {
        "sourceSystemCode": "enum | MBFALI, Boss, MBF",
        "sourceMembershipId": "string",
        "sourceSecondaryRefNum": "string",
        "sourceCurrARaccountId": "long (nullable)",
        "sourcePrevARaccountId": "long (nullable)",
        "bossMembershipNum": "long",
        "sourceUPI": "string",
        "bossSuffixNum": "int (nullable)"
    },
    "lastCardOrderDetail": {
        "lastCardOrderDate": "DateTime?",
        "numberOfCardsOrdered": "int?"
    },
    "paymentStatus": {
        "status": "MembershipPaymentStatus",
        "errorDescription": "string",
        "sessionId": "long",
        "contributorId": "long?",
        "minPaymentAmount": "decimal?",
        "maxPaymentAmount": "decimal?",
        "arrearsBalance": "decimal?",
        "arrearStatus": "string",
        "ddNextCycleDate": "DateTime?",
        "currentDPT": "DateTime?",
        "additionalResult": {
            "bossPaymentDate": "int",
            "bossRateDate": "int",
            "processSourceType": "string",
            "processSourceId": "string",
            "discountSourceType": "string",
            "discountSourceId": "string",
            "renewalNumber": "int",
            "renewalSlipNumber": "sbyte"
        }
    },
    "premiumCommunicationDetails": {
        "communicationId": "long",
```

```
        "packageDescription": "string",
        "paymentFrequency": "string",
        "phirFlag": "string",
        "preAprilPremiumDetail": {
            "cpiRebatePercentage": "decimal?",
            "netRate": "decimal?",
            "rebateAmount": "decimal?",
            "nonRebateRate": "decimal?",
            "rebateRate": "decimal?",
            "lhcLoad": "decimal?",
            "discountPercentage": "decimal?",
            "abdAmount": "decimal?"
        },
        "postAprilPremiumDetail": {
            "cpiRebatePercentage": "decimal?",
            "netRate": "decimal?",
            "rebateAmount": "decimal?",
            "nonRebateRate": "decimal?",
            "rebateRate": "decimal?",
            "lhcLoad": "decimal?",
            "discountPercentage": "decimal?",
            "abdAmount": "decimal?"
        },
        "netRatePercentageChange": "decimal?",
        "rebateAmountPercentageChange": "decimal?",
        "nonRebateRatePercentageChange": "decimal?",
        "rebateRatePercentageChange": "decimal?",
        "lhcLoadPercentageChange": "decimal?",
        "discountPercentageChange": "decimal?",
        "cpiRebatePercentageChange": "decimal?",
        "abdPercentageChange": "decimal?"
    },
    "emailMessage": [
        {
            "id": "long",
            "membershipId": "long",
            "personId": "long?",
            "dateSent": "DateTime",
            "sendTo": "string",
            "sendToEmail": "string",
            "sendToCcEmail": "string",
            "sendBy": "string",
            "subject": "string",
            "body": "string",
            "emailClientType": "int",
            "sendStatus": "string",
            "sendFailures": "long?",
            "auditTimeStamp": "DateTime",
            "attachments": [
                {
                    "id": "string",
                    "name": "string",
                    "data": "string"
                }
            ],
            "replyTo": "string"
        }
    ],
    "clearanceOutHistory": [
        {
            "transactionId": "string",
            "referenceId": "string",
            "channel": "string",
            "processDate": "DateTime?",
            "message": "string",
            "status": "string",
            "code": "string",
            "responseText": "string",
            "userId": "string"
        }
    ],
    "corporateGroupExitSummary": {
        "groupExitDate": "DateTime?",
        "groupName": "string",
        "orgId": "long?",
        "currentPackageName": "string",
        "enrolYear": "long?",
        "contributorAge": "long?"
    },
    "personProductHistory": [
        {
            "membershipId": "long",
            "suffix": "int",
            "productType": "string", // Comma-separated values: "Both, Hospital, Ancillary"
            "startDate": "DateTime?",
            "endDate": "DateTime?",
            "productOptId": "long",
```

```
            "gapProductFlag": "string",
            "gapProductOptId": "string",
            "gapFreeProductOptId": "string",
            "productRank": "int",
            "familyType": "string",
            "waitingPeriodGroup": "int",
            "benefitLevelPeriodGroup": "int",
            "benefitMaxPeriodGroup": "int",
            "fundNo": "int",
            "portFlag": "string",
            "newBornFlag": "string",
            "portedFundCode": "string",
            "portedProductCode": "string",
            "portedProductName": "string",
            "portedFundName": "string",
            "suspendedFlag": "string"
        }
    ],
    "eftMembershipDetails": {
        "bsbNumber": "string",
        "accNumber": "string",
        "amount": "string"
    },
    "membershipsAdditionalProperties": {
        "isAIC": "boolean",
        "isOSHC": "boolean",
        "isNonOSHCAG": "boolean",
        "isDDCopaid": "boolean",
        "isCeasedorFutureCeased": "boolean"
    },
    "corporateMembership": {
        "memberName": "string",
        "emailAddress": "string",
        "dateOfBirth": "string",
        "orgId": "string",
        "membershipNumber": "string",
        "contactId": "Guid",
        "createdDate": "DateTime",
        "coverStartDate": "DateTime"
    },
    "entitlements": [
        {
            "presentingIllnessCode": "string",
            "presentingItem": "string",
            "errorCode": "string",
            "errorMessage": "string",
            "hospitalEntitlements": {
                "assessCode": "int?",
                "assessText": "string",
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                "excessBonus": "decimal?",
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                "coPaymentAmountDescription": "string",
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                "claimGroupDescription": "string",
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                "packageCodeForCoveredAtProduct": "string",
                "waitingPeriods": [
                    {
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                        "duration": "string",
                        "description": "string",
                        "unit": "string",
                        "endDate": "DateTime?"
                    }
                ],
                "personExcesPaidYTD": "double",
                "memberExcesPaidYTD": "double"
            }
        }
    ],
    "eligibilityHistory": [
        {
            "oecId": "string",
```

```
            "receivedDate": "DateTime",
            "admissionDate": "DateTime",
            "isSameDay": "bool",
            "expectedLengthOfHospitalStay": "int",
            "peaRequestIndicator": "bool",
            "accidentIndicator": "bool",
            "membershipNumber": "long",
            "personId": "long",
            "suffixNumber": "int",
            "tableScale": "string",
            "datePaidto": "DateTime?",
            "financialStatus": "bool",
            "tableName": "string",
            "excessAmount": "decimal",
            "excessAmountDescription": "string",
            "excessBonusAmount": "decimal",
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            "oecAssessmentText": "string",
            "oecResponseCode": "string",
            "peaPotentialIndicator": "string",
            "errorMessage": "string",
            "hospitalProviderNumber": "string",
            "hospitalProviderName": "string"
        }
    ]
}
```

Upstream and Downstream Systems

- **Upstream systems:**

  Apollo
  Cyclops
  Batch Jobs
  Digital Cards
  Digital (MyBupa)
  Facade Services
  IVR
  SpeechStorm
  Titanium
  Corporate CPC
  Connexions CPC
  Access Verification
  Blua
  PACE ENE

- **Downstream systems:**

  BOSS

Caching

Caching Component Details

Caching Component - Redis

Integration Approach

How to integrate HI.Library.Cache component with domain API

Authentication

Authentication for the Membership Domain API is implemented using **Bearer Token Authentication** (OAuth 2.0 + JWT).

- **Client Applications** obtain an access token from **Microsoft Entra** using either:
  - **User Credentials** (Public Clients), or
  - **Managed Identity** (Confidential Clients).
- The access token **must** include the required permissions as **scopes** or **roles**.
- For every request to the Membership Domain API, the client application must send the access token as a **Bearer token** in the `Authorization` header:

```
Authorization: Bearer <access_token>
```

- The Bearer token is validated, and access to the requested endpoint is granted accordingly.

Additional References

- **Thick Client to Domain API (Cyclops to Domain API)**
  Thick Client to Domain API (Cyclops to Domain API)- Authentication and Authorization Flow
- **API to API Flow (BFF to Domain API)**
  Authentication and Authorization Flow – DotNet Team - Confluence

Token Validation Logic

The JWT is validated at the **API Management (APIM)** level for every request. Validation checks include:

- Valid Issuer
- Valid Audience
- Token Signature
- Token Lifetime

If any validation fails, the request is rejected with a **401 Unauthorized** response.

> Note: Similar token validation functionality exists within the `HI.Library.Authorization` package and can be enabled or disabled via configuration. Currently, this package's token validation is bypassed because tokens are validated at the APIM level.

Authorisation

Authorisation Overview

Authorisation is performed based on the **scopes** or **role claims** present in the JWT bearer token of each request. The `HI.Library.Authorization` package is utilized by the Membership Domain API to implement the authorisation logic.

How is Authorisation Implemented?

- All applicable permissions such as `MembershipDomainAPI.Create`, `MembershipDomainAPI.Read`, etc., will be initialized for authorisation purposes.

- Endpoints requiring authorisation will be annotated with the `[AuthorizePermission]` custom attribute, specifying the required permission as an argument.
  Example:

  ```
  [AuthorizePermission("MembershipDomainAPI.Read")]
  ```

  This would typically be used on an HTTP GET method.
  Refer to the complete list of permissions in the **Membership - API Design | Per Endpoint Access Rules** section.
- When a request is made to an endpoint, the authorisation handlers validate that the required permissions (specified in the `[AuthorizePermission]` attribute) exist within the `scope` or `roles` claim of the JWT.
- If the required claims are present and valid, access to the endpoint is granted. Otherwise, the request is rejected with a **401 Unauthorized** response.

Mapping of Users/Groups to Roles

> **Note:** This section applies to **Public Client Applications** only.

- For **Cyclops**, users will be added to the Azure Entra group:
  `RES-CLD-Cyclops-Hugo-dev-access`
  This group is associated with the App Registration during User Onboarding.
- Similarly, other public client applications (involving user interaction) can have their own groups, which can be assigned to their respective App Registrations.

Per-Endpoint Access Rules

- Each endpoint **must** use the `[AuthorizePermission]` custom attribute provided by the `HI.Library.Authorization` package.
- The following permissions apply to the Membership API Endpoints (list to be provided in the detailed API design documentation).

Permission	Applicable HTTP Verb
MembershipDomainAPI.Read	GET
MembershipDomainAPI.Create	POST
MembershipDomainAPI.Update	PUT
MembershipDomainAPI.Delete	DELETE

Logging

The **Logging component** provides a centralized logging solution for Hugo Domain APIs. It is implemented as a separate class library, enabling seamless integration across microservices as a reusable NuGet package.

- **Package Name:** `HI.Library.Logging`
- **Logging Design Documentation:**
  Logging - Design

Architecture & Implementation

- Logs are sent to **Azure Application Insights** for centralized monitoring and analysis.
- Uses the `ILogger` interface from **Microsoft.Extensions.Logging** to capture logs across Domain APIs.

- **OpenTelemetry** is leveraged to collect and export logs to Azure Application Insights via the `Azure.Monitor.OpenTelemetry.AspNetCore` package.

Key Logged Fields

Field Name	Description	Field Type
Timestamp	The date and time when the log entry was created.	Telemetry
Message	The main content of the log, describing the event or issue.	Telemetry
Severity Level	The severity of the log (e.g., Info(1), Warning(2), Error(3), Critical(4)).	Telemetry
Operation ID	A unique identifier for the API request to correlate logs across services.	Telemetry
Operation Parent Id	This field represents the ID of the parent operation. It is used to correlate nested operations, such as a request and its dependent calls.	Telemetry
Cloud Role Name	This field specifies the role name of the cloud service where the telemetry data is coming from.	Telemetry
Method	The HTTP method used for the request (e.g., GET, POST, PUT, DELETE).	Telemetry/ Custom
Path	This field represents the URL path of the request. It provides the endpoint that was accessed.	Telemetry/ Custom
Trace Id	This field represents a unique identifier for a single trace.	Telemetry
Item Type	This field specifies the type of telemetry item (e.g., request, exception, trace).	Telemetry
Session ID	The identifier for a user session, tracking the user's actions within a specific time frame.	Custom
Membership ID	The identifier associated with a user's membership	Custom
Customer ID	This field represents the unique identifier for a customer	Custom
CorrelationID	A custom Correlation ID used to trace and correlate requests and responses across different services and components within a distributed system.	Custom

Log levels and retention

Log Level	Description
Log Information	Provides informational messages about the normal operation of the application.
Log Warning	Highlights potential issues or unexpected behavior that does not disrupt the application.
Log Error	Captures application errors or exceptions that prevent a specific operation from succeeding.
Log Critical	Logs severe errors that cause the application or a critical component to fail.

Retention Policy:

- Application Insights retains log data for a configurable retention period (90 days).

Exception and Error Handling

Global Exception Handling Pattern

Exceptions are handled using the **Global Exception Handling Middleware** provided by the `HI.Library.ExceptionHandling` package.

Mapping Exception Types to HTTP Status Codes

Exception Type	HTTP Status Code	Description
Bad Request	400 Bad Request	The server could not understand the request due to invalid syntax.
Unauthorized	401 Unauthorized	Authentication is required and has failed or has not yet been provided.
Forbidden	403 Forbidden	The client does not have access rights to the content.
NotFound	404 Not Found	The server can't find the requested resource.
RequestTimeout	408 Request Timeout	The server timed out waiting for the request.
Conflict	409 Conflict	The request conflicts with the current state of the server.
PreconditionFailed	412 Precondition Failed	The server does not meet one of the preconditions in the request headers.
Unprocessable Request	422 Unprocessable Entity	The request was well-formed but unable to be followed due to semantic errors.
InternalServerError	500 Internal Server Error	The server has encountered a situation it doesn't know how to

		handle.
NotImplemented	501 Not Implemented	The request method is not supported by the server and cannot be handled.
BadGateway	502 Bad Gateway	The server received an invalid response from the upstream server.
ServiceUnavailable	503 Service Unavailable	The server is not ready to handle the request, often due to maintenance.
GatewayTimeout	504 Gateway Timeout	The server did not receive a timely response from an upstream server.

Custom error schema

```
{
  "data": {},
  "errors": [
    {
      "code": "string",
      "message": "string",
      "severity": "enum (Possible values: \"NotSet\", \"Information\", \"Warning\", \"Error\")",
      "classification": "enum (Possible values: \"Business\", \"System\")"
    }
  ]
}
```

Request Payload Storage in Application Insights

- When an exception occurs, the **request payload** is **encrypted** and stored in **Application Insights** as a property within the custom dimensions.
- Sensitive values are securely stored in **Azure Key Vault**.

Handling Custom Correlation Id (`X-Correlation-Request`)

- If a custom correlation ID is **not present** in the HTTP request header, the exception handling middleware **generates one** and assigns it to the header.
- If a correlation ID **already exists**, the middleware logs the exception in Application Insights using the **existing correlation ID**.

API Performance and Capacity Metrics

Metric Type	Metric	Value
Functional Load	Error rate [Failed requests per minute]	6/hour or .1/minute
	Availability [System uptime percentage]	100%
Latency	Average response time	300 ms
	90th percentile	1 sec

	Worst case	1.5 sec
Throughput	Average requests per minute	4000 calls/min
	Peak requests per minute	6000 calls/min
Rate limiting and throttling	Rate limited on the API Level	2000 requests per IP/min
Expected Resource Use	Limits: Memory	1Gi
	Requests: CPU	400m
	Requests: Memory	750Mi
	HPA Configuration: Min Replicas	6
	HPA Configuration: Max Replicas	10

Infrastructure and Deployment Design

At the API level, define

- AKS namespace and container info:
  - Namespace: **hicbocoreservices-{env}-ns**
  - Container: **hidomain-membership**
- Ingress/APIM setup
  - APIM API Name - **himembershipdomain/{env}**
- ConfigMap and Secret usage
  - Library Variable Group: **hicbocoreservices-membership-api-{env}-vars**
- Infrastructure CI/CD pipeline details:
  - **<To be Provided later>**
- Environment strategy (Dev, T27, Prod, etc.)
  - DEV: **d24**, SIT: **d34**, PERF: **t33**, UAT:**euat**, PROD: **prod**

Observability

AppDynamics/App Insights Integration Setup

- AppDynamics Tier/node mapping: NA
- AppDynamics Custom business transactions and metrics: NA
- App Insights: **hicbocoreservices-{env}-ai01**

AppDynamics/AppInsights Alerts

- CPU/memory/error rate thresholds: Api Alerts
- Notification channels: ServiceNow Ticket and Email Alert to dotnet support team.

Repository and Pipelines

Details	Link

Git repository	HI.API.Membership - Repos
Branching strategy	Branching Strategy for .NET based projects - DotNet Team - Confluence
Build and release pipelines	To be updated later
Security scans, tests, validations	To be updated later

Non-Functional Requirements

- Scalability: HPA should be enabled.
- Resilience and failover strategy (DR): API will be Active - Active running in both AKS clusters **productionaks** and **productiondraks.** For infrastructure DR refer the document:
  - Hugo Core Services and Batch Jobs Technical Recovery Procedure - Disaster Recovery
- Security and compliance standards (e.g., OWASP): SCR_008 API Security & Management

---

Secrets Management

- Use of Key Vault or secret manager: Key Vaut will be used **banz-hi-membership-{env}-kv01**
- Rotation and access control:
  - Manual yearly rotation of key vault entries has to be done for the following **<More details will be filled after the initial release>**

Key Vault Value	Description
ApimKeyValue	Subscription key for authenticating with Azure API Management (APIM).
ClientCertificateBase64	Base64-encoded client certificate for mutual TLS authentication.
CacheBlueKeyCPHSalt	Salt used in encryption of BlueKey CPH cache data.
CacheUnsubscribeCommPreferenceSalt	Salt used for unsubscribe communication preference cache encryption.
CacheBlueKeyCERTSalt	Salt used for encrypting BlueKey certificate cache entries.
CacheBluekeyRefDataSalt	Salt used for encrypting BlueKey reference data cache.
MembershipDomainSettings--ClientOptions--ClientCertificateBase64	Client certificate for secure Membership API access.
MembershipDomainSettings--ClientOptions--DefaultRequestHeaders--	APIM subscription key for Membership service calls.

Ocp-Apim-Subscription-Key	
InteractionDomainSettings--ClientOptions--ClientCertificateBase64	Client certificate for secure Interaction API access.
CrmApiSettings--ServiceUsers[1]--ClientOptions--JwtTokenOptions--Password	Client password for CRM API user 1.
CrmApiSettings--ServiceUsers[2]--ClientOptions--JwtTokenOptions--Password	Client password for CRM API user 2.
CrmApiSettingsPNP--ClientOptions--JwtTokenOptions--Password	Password for PNP CRM integration.
AzureMonitor--EncryptionKey	Encryption key for logs sent to Azure Monitor.
AzureMonitor--EncryptionIV	Initialization Vector for Azure Monitor encryption.
ServiceBus--EncryptionKey	Key used to encrypt messages on the Service Bus.
ServiceBus--EncryptionIV	Initialization Vector used with Service Bus encryption.
ProductCatalogueAPI--SubscriptionKey	Subscription key for accessing Product Catalogue API.
ProductCatalogueAPI--ClientSecret	Client secret for authenticating Product Catalogue client.
TokenSettings--clientSecret	OAuth client secret used with client ID.

API Specification

Income Tier

https://bupateam.atlassian.net/wiki/spaces/DT/folder/4280750223?atlOrigin=eyJpIjoiMDhkYmI0YmY1ZmJjNGJhMWFmOWFkNDQxZWYyY2NjNTMiLCJwIjoiYyJ9 Can't find link

Alert

CVO

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Membership

Person Covered

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Cover Details

Enrollment

https://bupateam.atlassian.net/wiki/spaces/DT/folder/4323640516?atlOrigin=eyJpIjoiZDQwZGYzM2IyZWI0NDE0NjlkZWZhZGEzMjljNjgyOTkiLCJwIjoiYyJ9 Can't find link

Tax Statement